

Coding Agents for Detecting and Refactoring Design and Architecture Smells

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Today's Presentation

- I will present my current studies, in collaboration with Dalhousie University:
 - **Detecting** Design and Architecture Smells
 - **Refactoring** Design and Architecture Smells

Detection Study

Using non-agentic and agentic approaches to detect Design and Architecture Smells

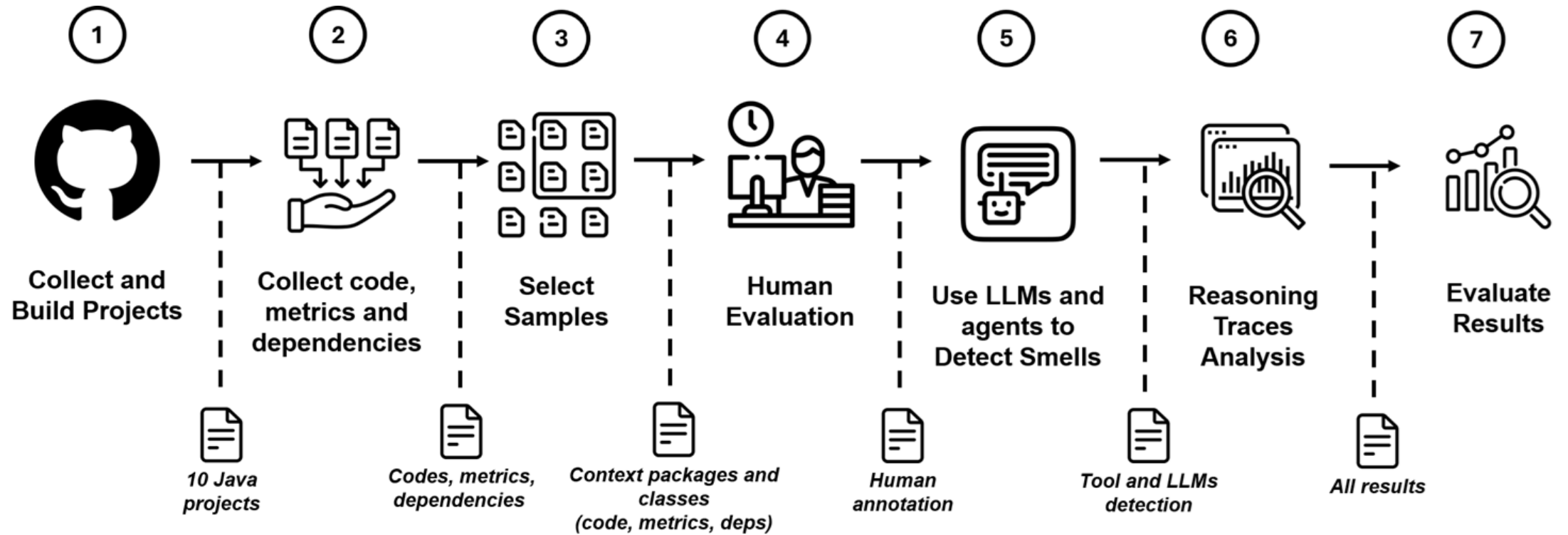
Goal

- Evaluate how different input contexts and agentic approaches affect smell detection at different software granularity levels.
- Target Smells:
 - Insufficient Modularization (IM)
 - Hub-like Modularization (HM)
 - God Component (GC)
 - Unstable Dependency (UD)

Our Questions

- RQ1. How aligned are **non-agentic LLMs** with human evaluation in detecting design and architecture smells?
- RQ2. How aligned are **coding agents** with human evaluation in detecting design and architecture smells?
- RQ3. How do different context strategies shape **reasoning traces** for design and architecture smell detection?

Study Steps



Results

Smell	Approach	F1	Kappa
God Component	Rawcode + GPT 5 mini	0,84	0,66
God Component	Rawcode + Kimi K3	0,85	0,65
Unstable Dependency	Metrics + deps + GPT 5 mini	0,77	0,73
Unstable Dependency	Metrics + deps + Kimi K3	0,75	0,71
Hub-like Modularization	Agentic + Codex (GPT 5.4 mini)	0,86	0,75
Hub-like Modularization	Agentic + mini-swe-agent (GPT)	0,83	0,70
Insufficient Modularization	Metrics + deps + Qwen Coder 3	0,71	0,52
Insufficient Modularization	Metrics + deps + GPT 5 mini	0,67	0,41

Finds

- RQ1. *“Non-agentic LLMs show meaningful alignment with human judgment [...]. Metrics and dependencies generally align better with human decisions than raw code.”*
- RQ2. *“Agentic tools show uneven alignment with human judgment across smells [...] agentic tools are competitive, however [for some smells] remain below the best non-agentic results.”*
- RQ3. *“Raw code lead to semantic, responsibility-based reasoning [...] Metrics and dependencies promote metric-driven and relational reasoning based on implicit thresholds [...] Coding agents combine both forms of evidence by inspecting repositories, extracting simple metrics, and interpreting class or package roles.”*

Current Status

- Adding a new model analysis (Kimi K3).
- Refining the paper: Refused in SCAM
- Submission Goal: SANER.

Refactoring Study

*Multi-agent Refactoring for Design and
Architecture Smells (MaRS)*

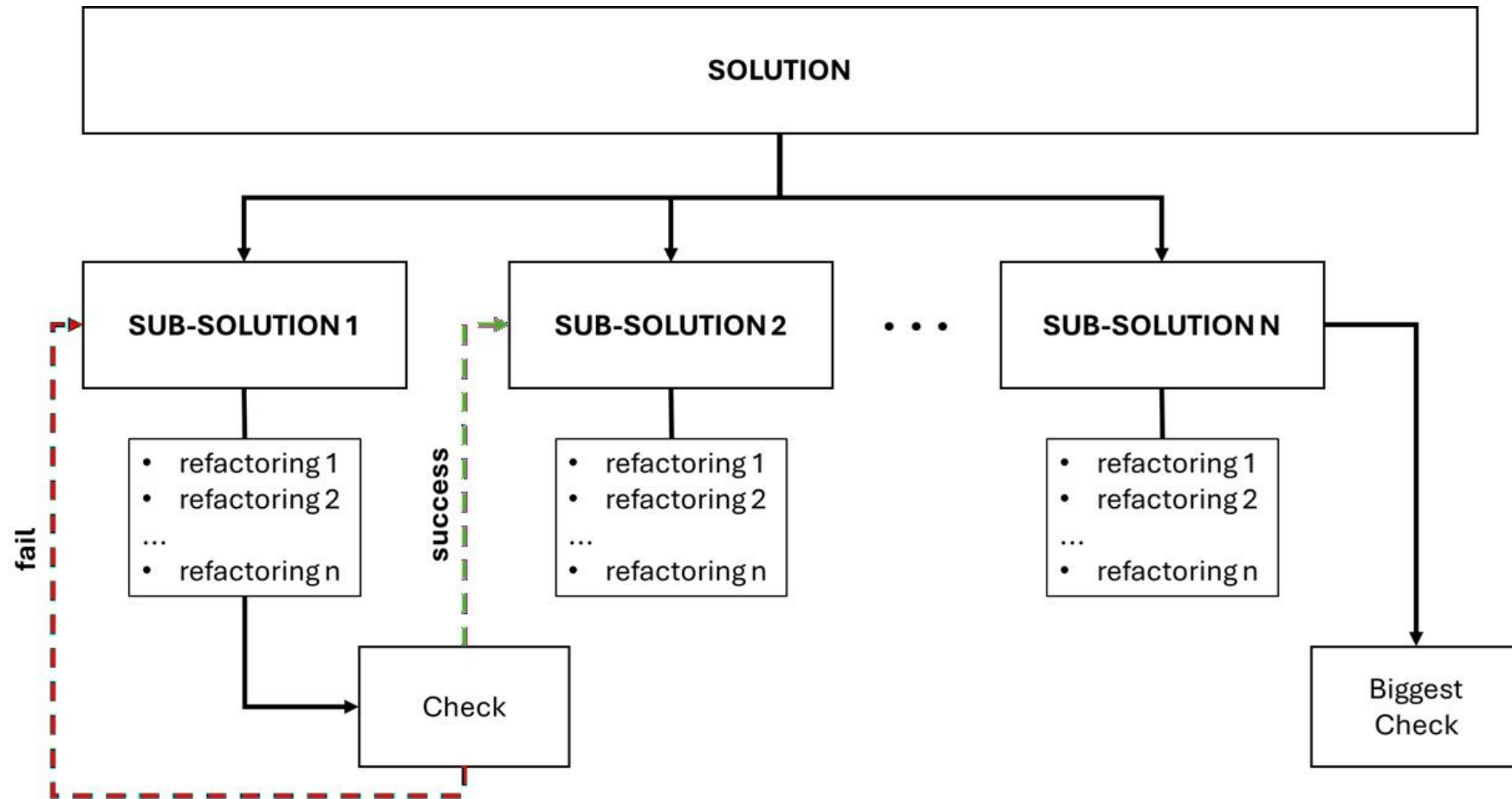
Goal

- Compare the capabilities of MaRS and Coding Agents in refactoring design and architecture smells.

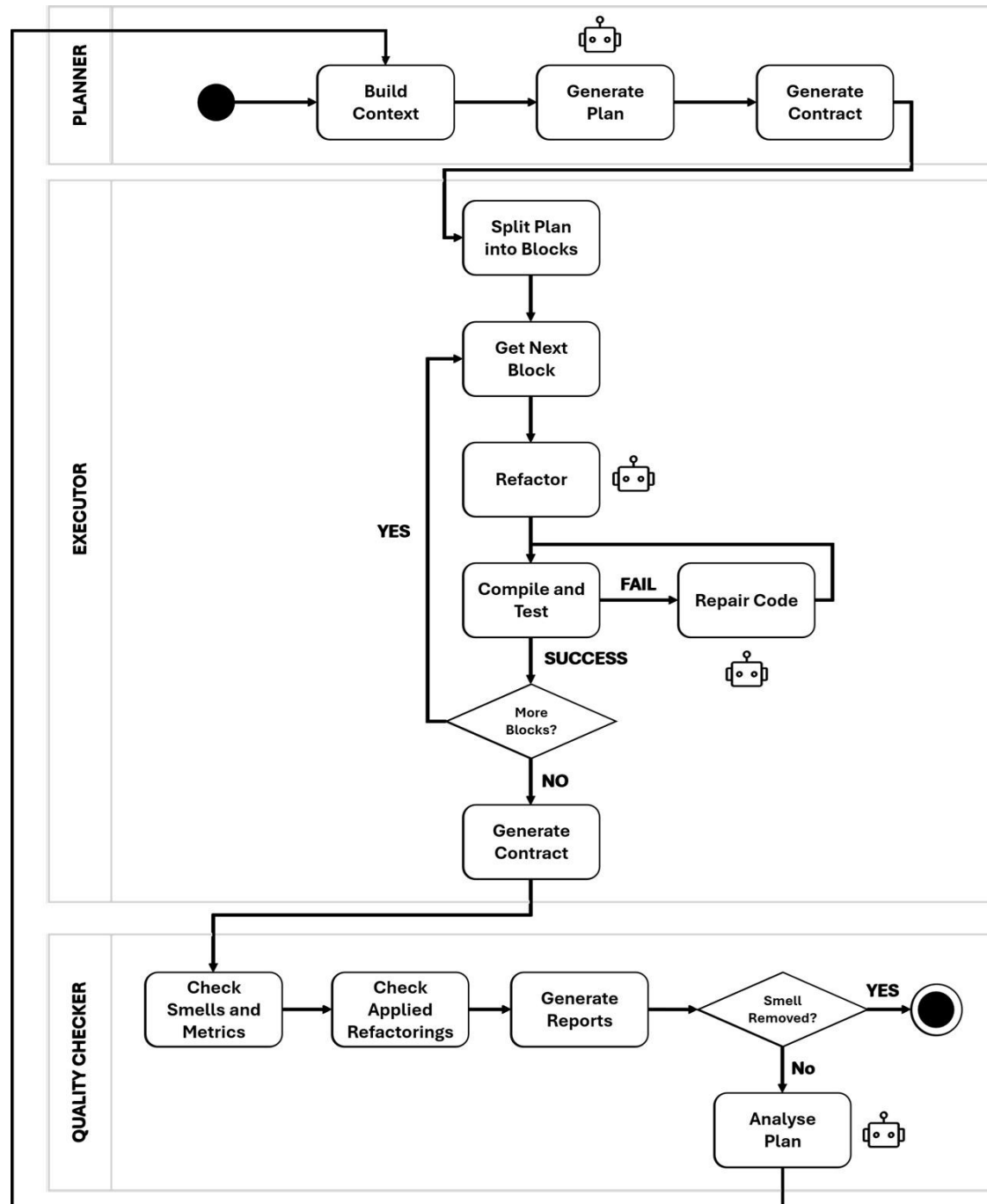
Our Questions

- RQ1. How effective are existing **coding agents** in improving design and architecture quality indicators?
- RQ2. How does **MaRS** compare with existing coding agents in improving design and architecture quality indicators?
- RQ3. What **refactoring patterns** characterize successful improvements in design and architecture quality indicators?
- RQ4. What are the main differences between refactoring done by **developers and models** for improving software design and architecture?

Our Approach



MaRS Workflow



MaRS Settings

```
project:
  name: commons-io
  repo_path: /data/henrique/multiagent_refactoring/data/repositories/commons-io

runs: 2

target:
  smell: GC
  smell_name: God Component
  target_type: package
  target_name: org.apache.commons.io.filefilter

planner:
  base_config: /data/henrique/multiagent_refactoring/mvp/planner/config.test.yml

source_refactor:
  base_config: /data/henrique/multiagent_refactoring/mvp/source_refactor/config.test.yml

quality_checker:
  base_config: /data/henrique/multiagent_refactoring/mvp/quality_checker/config.test.yml

output:
  runs_dir: /data/henrique/multiagent_refactoring/data/runs
```

MaRS Audition

```
"removed": [
  {
    "level": "design",
    "smell": "Insufficient Modularization",
    "entity": "org.apache.commons.io.input",
    "source": "DesignSmells.csv",
    "row": {
      "Project": "commons-io",
      "Package": "org.apache.commons.io.input",
      "Class": "Tailer",
      "Smell": "Insufficient Modularization",
      "Description": "The tool detected the smell in",
      "File": "/data/henrique/multiagent_refactoring",
    },
    "key": "design|Insufficient Modularization|org.a
  }
],
"added": []
```

```
"refactoringminer": {
  "enabled": true,
  "ok": true,
  "json_path": "/data/henrique/multiagent_refactoring/data",
  "log_path": "/data/henrique/multiagent_refactoring/data/",
  "refactorings_count": 19,
  "types": {
    "Move Method": 7,
    "Extract Class": 1,
    "Move Class": 2,
    "Change Class Access Modifier": 2,
    "Remove Class Modifier": 4,
    "Change Method Access Modifier": 2,
    "Change Return Type": 1
  }
},
```

Current Status

- MaRS is ready (except for UD).
- Creating dataset:
 - Collect 20 Java repositories (> 5k commits)
 - Run DesigniteJava on the commits from the last 5 years.
 - Check cases which IM, HM, GC and UD were removed.
 - Manual analysis to select samples.

Thank you!

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